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Letter

Perovskite Solar Cells Made by a Self-Quenching Method Using a Volatile Perovskite Ink with Safer Alternatives to 2-Methoxyethanol

Samy Almosni,* Fumiyasu Awai, Ajay Jena, Ludmila Cojocaru, Ryota Kan, Linh Vu Tuy, Keishi Tada, Kazuteru Nonomura, Jotaro Nakazaki, Satoshi Uchida, and Hiroshi Segawa*

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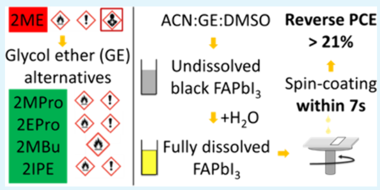
ABSTRACT: Four solvents from the glycol ether (GE) family were evaluated as safer alternatives to the highly toxic 2-methoxyethanol (2ME) used for the fabrication of perovskite solar cells (PSCs). A self-quenching process by spin-coating the acetonitrile (ACN)-based ink for 7 s was developed. The influence of the ACN:GE ratio on the coverage of perovskite layers and cell efficiency was investigated to find that with the ACN:GE ratio of 8:1 the four solvents studied here can produce PSCs with an efficiency similar to 2ME and therefore can be used as a replacement for the latter.



KEYWORDS: *Perovskite Solar Cell, Solvent Toxicity, Volatile Ink, Glycol Ethers, Self-quenching*

Lead halide perovskite is gaining a lot of interest for next-generation solar cells thanks to the low purity and amount of materials needed to fabricate the devices, the possibility of processing them at low temperatures on flexible substrates, and their high-power conversion efficiency under different illumination conditions. These features demonstrate their potential low cost and high power-to-weight ratio. In 2024, efficiency over 26% under AM1.5G illumination has been certified for perovskite solar cells (PSCs).[1] Moreover, much progress has been made on encapsulation to improve their stability.[2] These results substantiate their potential to become the next generation of photovoltaic devices in use and explain why several industries worldwide are now investing in this field and are starting to build their pilot-scale and full-scale fabrication lines.[3] Among the approaches that have been developed to scale up perovskite solar cells, the solution process using fast-evaporating volatile solvents has shown promising results.[4−10] One of the challenges of this approach is to dissolve the perovskite precursor in a low boiling point and a low coordinating solvent such as acetonitrile (ACN). Several groups have reported the use of methylamine gas (MA) to dissolve $MAPbI_3$ in ACN and form a compact layer without any quenching and even without annealing.[11−14] However, use of MA gas in FA-based perovskites prevents formation of the thermally stable $FAPbI_3$ because of its reaction with FAI and formation of MAFA intermediate.[15,16] By using 2-methoxyethanol (2ME) as the main solvent to dissolve the perovskite precursor, $FAPbI_3$ has been successfully deposited, and efficiencies over 22% on the cell and module levels have been achieved using scalable techniques such as slot-die coating.[9,17] However, due to the lower nucleation rate of 2ME inks compared to ACN:MA-based inks, one of the drawbacks of the 2ME approach is the necessity of gas quenching to form a compact layer. The lower nucleation rate in this case is due to the lower vapor pressure and higher coordination of 2ME and the amount necessary to dissolve perovskite precursor compared to the ACN:MA approach. Moreover, the toxicity of 2ME might prevent its industrial use as the industry is shifting toward safer solvents for printed electronics.[18] It has been reported that 2ME is a teratogen in animals. For example, in rabbits, an increase of 11% in embryonic death was reported when pregnant females were exposed by inhalation to only 10 ppm for 6 h per day during the gestation days 6−18.[19] Hence, for the safety of the workers using such recipes in research or industrial environments, we believe that the use of 2ME should be avoided if alternatives with similar functionalities can be found.

In this paper, we propose four alternatives to 2ME, namely, 1-methoxy-2-propanol (2MPro), 1-ethoxy-2-propanol (2EPro), 1-methoxy-2-butanol (2MBu), and 2-isopropoxyethanol (2IPE). To the best of our knowledge, the use of these solvents has never been reported in peer-reviewed articles to fabricate perovskite solar cells. These alternatives have been chosen because they belong to the same solvent family as 2ME, i.e., glycol ethers (GE), have boiling points close to that of 2ME and do not have the carcinogenic/reprotoxic pictogram, manifesting that they can be safer alternatives than 2ME.[20] The lower toxicity of 2MPro, 2EPro, and 2IPE compared to 2ME is also confirmed (Table S1) as they have a 250-fold higher threshold limit value (TLV), a level of occupational exposure to a dangerous substance where it is believed that healthy

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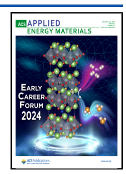

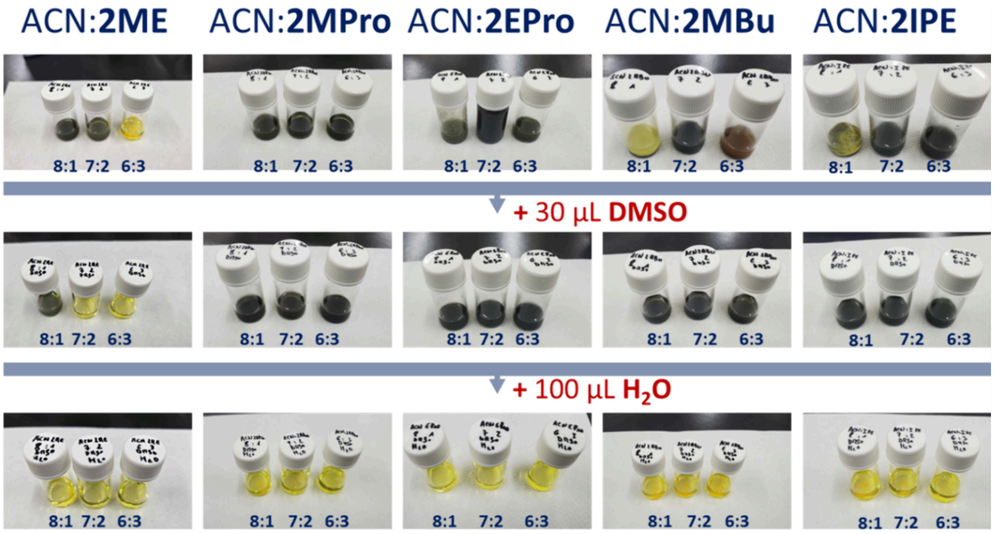

Figure 1. Perovskite ink prepared with different GEs and varying ACN:GE ratios before and after adding DMSO and water.

workers can work safely, than 2ME (no data could be found for 2MBu). TLVs are determined by the American Conference of Governmental Industrial Hygienists (ACGIH), and are recommended by the Occupational Safety and Health Administration (OSHA).[21] Hence, the above toxicity data clearly show that the proposed GEs in this study are potentially less toxic alternatives to 2ME. However, their use in the fabrication of PSCs has not been explored yet.

To quickly compare these four alternatives to 2ME and accelerate the research on volatile ink that can dissolve and form layers of $FAPbI_3$, we deemed the necessity to develop a spin coating recipe that could form a compact perovskite layer in air without an additional quenching method. Our approach was to use ACN-rich ink in order to increase the nucleation rate like the case of MA:ACN ink as these inks can form a compact layer without quenching. Due to its large impact on the precursor solubility and layer morphology, we first studied the effect of the ACN:GE ratio and water on the ink solubility.[9] Water was used due to its interesting coordination properties and high vapor pressure.[22] We then compared the influence of the GEs on the device efficiency and the layer's crystallographic and optoelectronic properties.[23]

We first assessed the coordination ability of these solvents using a MACl-doped $FAPbI_3$ perovskite and a solvent mixture composed of the different GEs, acetonitrile (ACN), dimethyl sulfoxide (DMSO), and water. 30 μL of DMSO was used to improve the crystallization as previously reported.[5] We found that the addition of 100 μL of water increases the solubility of the perovskite precursor in ACN-rich ink, without leading to fast precipitation of the solution or slowing down the crystallization too much as it would happen if we increased the DMSO amount to improve the solubility.[5,12] We found that all the selected GEs have weaker coordination to PbI_2 than 2ME. This was evident (Figure 1) from the presence of black perovskite precipitates in the vial after the addition of 30 μL/mL DMSO to all of the ACN:GE ratios (8:1, 7:2, 6:3) tested in this study. In other words, the solvent that dissolves the precursors well, without forming any black precipitate, displays a strong coordination. Hence, as can be seen in Figure 1, in the case of 2ME, only the 8:1 ratio shows a black precipitate, whereas 2MPro, 2EPro, 2MBu, and 2IPE formed a black precipitate for all the varied ratios. However, we found that adding 100 μL/mL of water on top of DMSO to these solvent systems could dissolve the perovskite precipitate and form a clear yellow solution with a fully dissolved precursor (Figure 1).

The prepared solutions were used to make films through a self-quenching method, that is without using any external air quenching, which is usually required for such GE-based recipes.[5,7,9] The modified solvent mixture in our case allowed us to make the films by spin-coating for only 7 s at a maximum spinning rate of 3000 rpm, without the necessity of external quenching. The layers were quenched by the airflow generated by the substrate spinning.[24−26] A clear effect of the ACN:GE ratio on the perovskite coverage and the quality of the perovskite layer was observed (Figure 2). We found that a

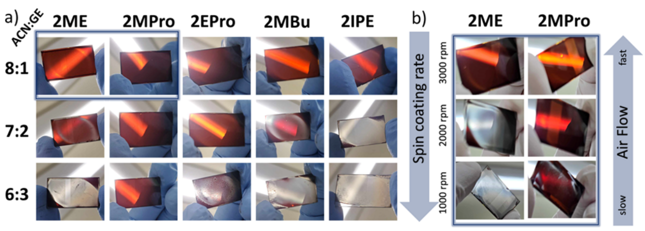

Figure 2. (a) Perovskite layers with different ACN:GE ratios and (b) perovskite layers with ACN:GE ratio of 8:1, coated at different spinning rates.

compact layer could be formed without external quenching by increasing the ACN content in these solvent systems. We assume it is due to the increased nucleation rate caused by the lower coordination and boiling point of ACN compared to those of GE solvents. As can be seen in Figure 2a, a high amount of GE generally leads to grayish layers, indicating incomplete coverage, seen in scanning electron microscopy (SEM) images presented in Figure S1. Interestingly, this effect of the ACN:GE ratio on film quality differed largely for the five GEs. From Figure 2a, it was clear that the solvent that shows the largest tolerance to the ACN:GE in terms of coverage is 2MPro followed by 2EPro, 2ME, 2MBu, and IPE. These differences can be explained by interplay between the coordination ability and boiling point of these solvents that eventually affect the crystallization and nucleation rate of the

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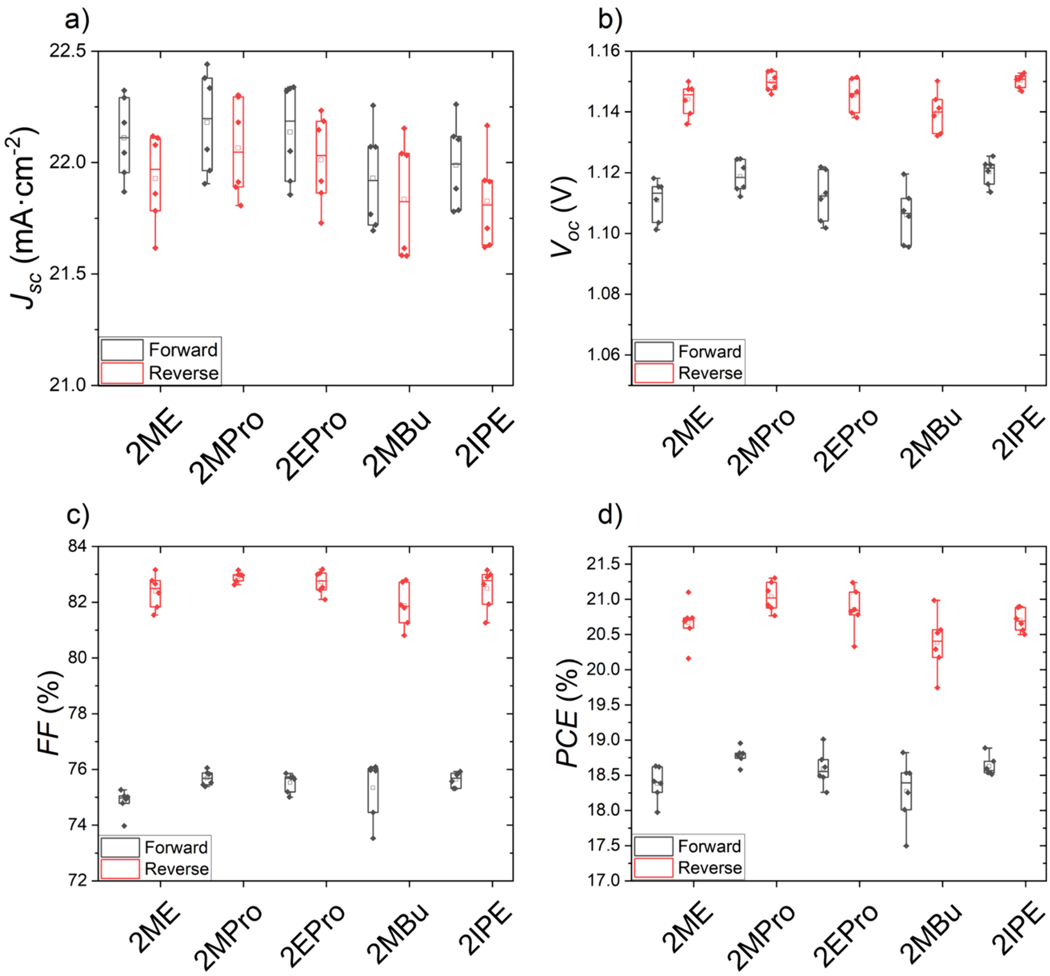

Figure 3. (a−d) I−V parameters of the perovskite solar cells prepared using perovskite ink including different GEs with a fixed ACN:GE ratio of 8:1.

perovskite layer.[5] From the observed difference in tolerance to the ACN:GE ratio, we expected a larger processing window, i.e., a larger tolerance to the airflow necessary to quench the layer, for 2MPro compared to 2ME. Since the airflow generated by the spinning substrate depends on the spinning rate, we spin-coated the 2ME and 2MPro inks with the ACN:GE ratio of 8:1 at different spinning rates; changing from 3000 to 1000 rpm to further reduce the self-quenching effect. From the picture presented in Figure 2b, 2MPro could be inferred to offer a larger processing window as compared to 2ME as it could still form a compact layer at a spinning rate as low as 1000 rpm, meaning a slower air flow is sufficient to quench the layer.

Since the 8:1 ACN:GE ratio and a spinning rate of 3000 rpm produced a compact perovskite layer with all five different GEs studied here, we used this solvent ratio and spinning rate for our further study. This process allowed us to fabricate solar cells with the widely studied planar n-i-p architecture employing SnO_2 and spiro-OMeTAD as the electron and hole transport layer, respectively (i.e., FTO/SnO_2/$FAPbI_3$/spiro-OMeTAD/Au). Figure 3 presents the light I−V parameters for the solar cells, using five GEs as precursor solvents for the fabrication of the perovskite layer. The best performances achieved with the systems descend in order 2MPro > 2EPro > 2IPE > 2ME > 2MBu with average reverse scan power conversion efficiency (PCE) of 21.03, 20.86, 20.70, 20.66, and 20.37%, respectively. Considering that the entire process is not fully optimized yet, fabricated below 120 °C, with relatively short annealing time and no chemical bath treatment of the SnO_2, we believe these efficiencies are promising although the values are slightly lower than that expected for $FAPbI_3$. The PCE of the cells was limited by the lower short circuit current density (J_{sc}) compared to state-of-the-art $FAPbI_3$ cells, and the cells suffer from hysteresis as confirmed, respectively, by the EQE (Figure S2) and I−V curves (Figure S3). This could be due to the presence of different crystal orientation, nonradiative recombination, or unoptimized band alignment between the ETL, perovskite, and HTL.[27] Nevertheless, these cells present encouraging shelf stability, keeping over 95% of their initial efficiency, after being stored in the dark in a dry room with a dew point of −35 °C, as shown in Figure S4. To understand if these solvents have a different impact on the perovskite crystal structure, morphology and optoelectronic properties the films were characterized by scanning electron microscopy (SEM), X-ray diffraction (XRD), time-resolved (tr), and steady-state (ss) photoluminescence (PL) measurements.

From the SEM cross-section images, presented in Figure 4a−e, it was apparent that all of the layers had a thickness of approximately 500 nm. The grains were mainly monolithic and grown vertically. From the top-view SEM image presented in Figure 4f−j, no major differences were seen between the five

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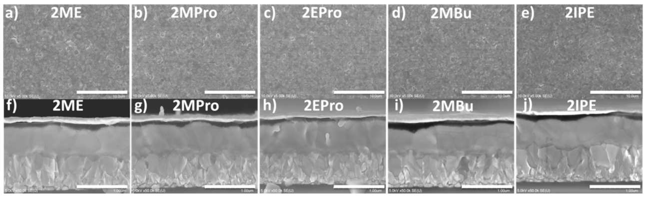

Figure 4. (a−e) Top-view SEM images of the perovskite layers (scale bar represents 10 μm) and (f−j) cross-section SEM images of the perovskite solar cells (scale bar represents 1 μm), with perovskite layers prepared using different GEs with a fixed ACN:GE ratio of 8:1.

GEs. Hence, in terms of grain morphology, all the films showed full coverage with small grain size as typically observed with layers made using the ACN:2ME solvent system.

The impact of the different GEs on the perovskite crystallinity was assessed by XRD. From the measurement performed on the samples with the ACN:GE ratio 8:1, presented in Figure S5, we could not identify any significant impact of the solvents on the perovskite crystallinity, crystal orientation, and strain. All of the samples showed a similar peak intensity ratio between the different crystal planes (Figure S5). Also, according to the Williamson Hall analysis, performed on these samples, they all showed values of microstrain between 0.13 and 0.16% and crystallite size between 80 and 110 nm.[28]

The effect of the different GEs on the perovskite optoelectronic properties was assessed by steady-state and time-resolved PL measurements performed on samples deposited on FTO/SnO_2 and on a glass substrate with and without spiro-OMeTAD layers. No appreciable impact could be identified from the results obtained from the samples with an ACN:GE ratio of 8:1, depicted in Figure S6. For the samples having the same architecture, the PL peak position varied between 1.53 and 1.55 eV. The peak full width at half maximum (FWHM) diverged by only 3 meV, and the carrier lifetimes were between 10 and 45 ns for the short lifetime and between 200 and 250 ns for long lifetime. These values are comparable to that reported by other work using 2ME as solvent and nitrogen quenching as the method of film formation.[29]

As for the difference observed in PCEs, which can be ascribed to slight differences observed in crystallinity, morphology, and PL, we believe that the outcome is more related to the fine-tuning of the deposition and annealing conditions than to the intrinsic nature of the solvents. In other words, as these solvents have different boiling points and surface tensions (Table S1), the optimal annealing temperature and spin-coating conditions to achieve an optimal crystal structure, morphology, and layer thickness can differ slightly for each of the GEs studied here. Nevertheless, the results manifest that the four GEs proposed here can be used as alternatives to 2ME to tailor the ink's chemical and physical properties such as coordination, vapor pressure, or surface tension, depending on the deposition technique used, to produce equally high-efficiency PSCs.

In conclusion, we assessed four unexplored safer alternative solvents for toxic 2-methoxyethanol, which is currently used in fast-evaporation methods for PSC fabrication via a self-quenching spin-coating process. Our results show that these solvents can be used to make perovskite layers with similar morphology, crystal structure, and optoelectronic properties as 2ME. Especially, 1-methoxy-2-propanol and 1-ethoxy-2-propanol seem to be promising alternatives as they show larger processing window and can produce PSCs with efficiencies of over 21% comparable to that of 2ME.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaem.4c02655.

Experimental section: Chemicals and materials, layers and solar cell fabrication, and GE properties. Characterizations: Light I−V, EQE, PL spectra, and X-ray diffractogram (PDF)

Movie S1: Video of spin-coating process (AVI)

■ AUTHOR INFORMATION

Corresponding Authors

Samy Almosni − *Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan;* orcid.org/0000-0001-8249-120X; Email: samy-almosni@g.ecc.u-tokyo.ac.jp

Hiroshi Segawa − *Department of General Systems Studies, Graduate School of Arts and Sciences, Komaba Institute for Science, Graduate School of Arts and Sciences, and Komaba Organization for Educational Excellence (KOMEX), College of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan; Research Center for Advanced Science and Technology (RCAST), The University of Tokyo, Tokyo 153-8904, Japan;* orcid.org/0000-0001-8076-9722; Email: csegawa@mail.ecc.u-tokyo.ac.jp

Authors

Fumiyasu Awai − *Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan;* orcid.org/0000-0001-5091-4693

Ajay Jena − *Komaba Institute for Science, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan*

Ludmila Cojocaru − *Komaba Institute for Science, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan;* orcid.org/0000-0002-3618-4142

Ryota Kan − *Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan*

Linh Vu Tuy − *Research Center for Advanced Science and Technology (RCAST), The University of Tokyo, Tokyo 153-8904, Japan*

Keishi Tada − *Department of General Systems Studies, Graduate School of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan;* orcid.org/0000-0002-3014-2515

Kazuteru Nonomura − *Komaba Organization for Educational Excellence (KOMEX), College of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan*

Jotaro Nakazaki − *Komaba Organization for Educational Excellence (KOMEX), College of Arts and Sciences, The University of Tokyo, Tokyo 153-8902, Japan;* orcid.org/0000-0002-2528-616X

Satoshi Uchida − *Research Center for Advanced Science and Technology (RCAST), The University of Tokyo, Tokyo 153-8904, Japan;* orcid.org/0000-0002-4971-574X

Complete contact information is available at:

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Notes

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